

Lecture 2

Polynomials over Finite Fields: Interpolation, Identity Testing, and the NTT

SEC 595 — Encrypted Computing

September 6, 2026

Roadmap

Lecture 1 built $\mathbb{Z}_q$. This lecture builds $\mathbb{F}_p[X]$, the polynomial ring over a finite field, and extracts from it four tools that between them account for most of the course.

The organising idea is a single duality. A polynomial of degree less than k can be written down in two completely equivalent ways:

$$\underbrace{(a_0, a_1, \dots, a_{k-1})}_{\text{coefficients}} \iff \underbrace{(a(x_1), a(x_2), \dots, a(x_k))}_{\text{evaluations at } k \text{ distinct points}}$$

Moving between these two representations is *interpolation*, and almost everything in this lecture is an application of being able to move between them cheaply.

Throughout, p is prime and $\mathbb{F}_p = \mathbb{Z}_p$ is the field from Lecture 1. Most worked examples use $\mathbb{F}_{17}$, chosen because $17 - 1 = 16 = 2^4$ makes it the smallest field in which the NTT is interesting.

1. The Ring $\mathbb{F}_p[X]$

1.1 Degree and Division

$\mathbb{F}_p[X]$ is the set of polynomials $a(X) = a_0 + a_1X + \dots + a_dX^d$ with $a_i \in \mathbb{F}_p$. It is a commutative ring under the usual addition and multiplication. We write $\deg a$ for the degree, with the convention $\deg 0 = -\infty$.

Theorem 1 (Division algorithm). *For $a, b \in \mathbb{F}_p[X]$ with $b \neq 0$ there exist unique $q, r \in \mathbb{F}_p[X]$ with*

$$a = q \cdot b + r, \quad \deg r < \deg b.$$

Compare this with the division algorithm for $\mathbb{Z}$ in Lecture 1: the role played there by “ $0 \leq r < n$ ” is played here by “ $\deg r < \deg b$ ”. That single analogy is enough to carry the whole Euclidean apparatus across.

Corollary 1. *$\mathbb{F}_p[X]$ is a Euclidean domain. In particular the Euclidean algorithm computes $\gcd(a, b)$, the extended version produces u, v with $ua + vb = \gcd(a, b)$, and an element is invertible modulo f exactly when it is coprime to f .*

Forward reference. This corollary is what will let us invert elements of $R_q = \mathbb{Z}_q[X]/(X^n + 1)$ in Lecture 3, and it is the algorithm behind the Li-Micciancio recovery attack on CKKS. Nothing new has to be learned; it is Lecture 1 with $\deg$ in place of $|\cdot|$.

1.2 Roots

Theorem 2 (Factor theorem). $a(\alpha) = 0$ if and only if $(X - \alpha) \mid a(X)$.

Proof. Divide: $a(X) = q(X)(X - \alpha) + r$ with $\deg r < 1$, so r is a constant. Evaluating at α gives $r = a(\alpha)$. $\square$

Corollary 2 (The key counting fact). *A nonzero polynomial of degree d over a field has at most d roots.*

Remark. “Over a field” is essential. In $\mathbb{Z}_8$, the polynomial $X^2 - 1$ has four roots: 1, 3, 5, 7. The proof of Corollary 2 needs the absence of zero divisors, and this failure is not a curiosity — it is exactly what makes arithmetic in R_q subtler than arithmetic in $\mathbb{F}_p[X]$.

Corollary 2 is doing more work in this course than any other single fact. Two polynomials of degree $< k$ that agree at k points are *equal*, because their difference has degree $< k$ and k roots. Everything in Sections 2 and 3 follows from that observation.

2. Interpolation: Coefficients $\leftrightarrow$ Evaluations

2.1 Lagrange

Theorem 3 (Lagrange interpolation). *Let $x_1, \dots, x_k \in \mathbb{F}_p$ be distinct and $y_1, \dots, y_k \in \mathbb{F}_p$ arbitrary. There is exactly one polynomial a of degree $< k$ with $a(x_i) = y_i$ for all i , namely*

$$a(X) = \sum_{i=1}^k y_i \ell_i(X), \quad \ell_i(X) = \prod_{\substack{j=1 \\ j \neq i}}^k \frac{X - x_j}{x_i - x_j}.$$

Each ℓ_i is the polynomial that is 1 at x_i and 0 at every other node, so the sum clearly interpolates. Uniqueness is Corollary 2.

Example 1. Let $x_1 = 1$, $x_2 = 2$, $x_3 = 3$, $y_1 = 10$, $y_2 = 5$, and $y_3 = 1$. Then,

$$\ell_1 = \frac{(X-2)(X-3)}{2}, \quad \ell_2 = \frac{(X-1)(X-3)}{-1}, \quad \ell_3 = \frac{(X-1)(X-2)}{2}$$

Plug x_i and each x_j into ℓ_i .

Example 2 (Reconstructing a Shamir secret in $\mathbb{F}_{17}$). This is the example from Lecture 15. The dealer picks $a(X) = 13 + 10X + 2X^2$ over $\mathbb{F}_{17}$ and hands out $a(i)$ for $i = 1, \dots, 5$:

$$(a(1), \dots, a(5)) = (8, 7, 10, 0, 11),$$

keeping the secret $S = a(0) = 13$. Any three shares recover it. Using nodes 1, 2, 3, and evaluating the Lagrange basis at $X = 0$:

$$\begin{aligned} \ell_1(0) &= \frac{(0-2)(0-3)}{(1-2)(1-3)} = \frac{6}{2} = 3, \\ \ell_2(0) &= \frac{(0-1)(0-3)}{(2-1)(2-3)} = \frac{3}{-1} = -3, \\ \ell_3(0) &= \frac{(0-1)(0-2)}{(3-1)(3-2)} = \frac{2}{2} = 1. \end{aligned}$$

Hence

$$S = 3 \cdot 8 + (-3) \cdot 7 + 1 \cdot 10 = 24 - 21 + 10 = 13. \checkmark$$

Two shares determine nothing: for *any* candidate secret s there is a unique degree-2 polynomial through the two known points and $(0, s)$.

2.2 Encoding a Set by Its Roots

A second way to hide information in a polynomial: put it in the *roots*.

Example 3 (Private set intersection, Lecture 16). Alice holds $\{3, 4, 5\}$ and forms

$$P(X) = (3 - X)(4 - X)(5 - X) = 60 - 47X + 12X^2 - X^3,$$

so that $P(y) = 0$ exactly when $y \in \{3, 4, 5\}$. She sends the encrypted coefficients. Bob, holding y , evaluates P homomorphically and returns $\text{Enc}(r \cdot P(y) + y)$ for random r . If y is in Alice's set the answer is y ; otherwise $r \cdot P(y)$ is uniformly random and reveals nothing.

The soundness of that protocol is again Corollary 2: a degree-3 polynomial cannot vanish at more than 3 points, so Bob cannot get a false “match”.

3. Polynomial Identity Testing

Suppose we want to check whether two polynomials A and B of degree $< d$ are equal, but they are enormous — say $d = 2^{20}$ — and we want to spend $O(1)$ work. Evaluate both at a single random point.

Theorem 4 (Schwartz–Zippel, univariate case). *Let $P \in \mathbb{F}[X]$ be nonzero of degree $\leq d$, and let $S \subseteq \mathbb{F}$ be finite. Then*

$$\Pr_{r \leftarrow S} [P(r) = 0] \leq \frac{d}{|S|}.$$

Proof. P has at most d roots by Corollary 2, and r is uniform on S . □

The multivariate version, which is what real proof systems use, replaces d by the total degree; the statement and the bound are otherwise identical.

Corollary 3 (The whole idea of a SNARK, in one line). *To check $A = B$, sample r uniformly and check $A(r) = B(r)$. If $A \neq B$ this fails to detect the difference with probability at most $d/|\mathbb{F}|$.*

Example 4. With $d = 2^{20}$ and $|\mathbb{F}| \approx 2^{254}$ (a typical SNARK field), the soundness error is about 2^{-234} . One field element, checked once, stands in for a million coefficient comparisons.

4. Vanishing Polynomials and Quotients

Let $H \subseteq \mathbb{F}_p^\times$ be a subgroup of order N — concretely, the set of N -th roots of unity.

Definition 1. The **vanishing polynomial** of H is $Z_H(X) = \prod_{h \in H} (X - h)$.

Proposition 1. *If H is the group of N -th roots of unity then $Z_H(X) = X^N - 1$.*

Proof. Every $h \in H$ satisfies $h^N = 1$, so each is a root of $X^N - 1$. There are N of them, all distinct, and $X^N - 1$ is monic of degree N ; by Corollary 2 there are no others. □

This is a remarkable convenience: the vanishing polynomial of a set of size N normally has N coefficients, but on a multiplicative subgroup it has two, and it can be evaluated with one exponentiation.

Theorem 5. *P vanishes on all of H if and only if $Z_H \mid P$, i.e. if and only if there is a quotient Q with $P(X) = Q(X) \cdot Z_H(X)$.*

Example 5 ($\mathbb{F}_{17}$, $N = 4$). Take $\omega = 13$, which has order 4 modulo 17 (Section 5). Then

$$H = \{1, 13, 16, 4\}, \quad Z_H(X) = X^4 - 1 = (X - 1)(X - 13)(X - 16)(X - 4) \text{ over } \mathbb{F}_{17}.$$

As a check, the roots sum to $1 + 13 + 16 + 4 = 34 \equiv 0$, matching the vanishing X^3 coefficient, and their product is $832 \equiv -1$, matching the constant term.

4.1 The Arithmetisation Pattern

Theorem 5 is the shape of every modern proof system. A computation is encoded as a constraint that must hold at every step; the steps are indexed by H ; and “the constraint holds at every step” becomes “the constraint polynomial $C(X)$ vanishes on H ”, which becomes

$$C(X) = Q(X) \cdot Z_H(X) \text{ for some polynomial } Q.$$

The prover supplies Q ; the verifier picks a random r and checks $C(r) = Q(r) \cdot (r^N - 1)$, which is Schwartz–Zippel again. A claim about a long computation has become one field-element identity.

5. Roots of Unity in $\mathbb{F}_p$

Everything in Sections 4 and 6 needs a subgroup of a prescribed size. Lecture 1 tells us exactly when one exists.

Proposition 2. *$\mathbb{F}_p^\times$ is cyclic of order $p - 1$, and contains an element of order N if and only if $N \mid p - 1$. Such an element is a **primitive N -th root of unity**, usually written ω .*

Example 6 ($\mathbb{F}_{17}$). 3 is a primitive root: $3^8 = 16 \equiv -1$, so $\text{ord}(3) = 16$. Therefore

$$\omega_{16} = 3, \quad \omega_8 = 3^2 = 9, \quad \omega_4 = 3^4 = 13, \quad \omega_2 = 3^8 = 16 = -1.$$

Definition 2. The **two-adicity** of p is the largest v with $2^v \mid p - 1$. It is the largest power-of-two NTT the field supports.

Field	$p - 1$	2-adicity	Used by
$p = 17$	2^4	4	our examples
$p = 3329$	$2^8 \cdot 13$	8	ML-KEM
$p = 8380417$	$2^{13} \cdot 1023$	13	ML-DSA
$p = 2^{64} - 2^{32} + 1$	$2^{32}(2^{32} - 1)$	32	STARK provers

Forward reference. Read the second and third rows carefully, because they decide the shape of the NTT in the two standards. Both use ring dimension $n = 256$. Splitting $\mathbb{Z}_q[X]/(X^n + 1)$ all the way into linear factors needs a primitive $2n = 512$ -th root of unity, hence $512 \mid q - 1$. ML-DSA has two-adicity 13, so it does. ML-KEM has two-adicity 8, so $256 \mid q - 1$ but $512 \nmid q - 1$: its transform stops one level early, at 128 quadratic factors. That is what “incomplete NTT” means, and we will do the negacyclic construction properly in Lecture 3.

6. The Number-Theoretic Transform

6.1 Definition

Fix $N \mid p-1$ and a primitive N -th root of unity ω . For $a = (a_0, \dots, a_{N-1})$, the transform is evaluation on H :

$$\text{NTT}(a)_j = \hat{a}_j = a(\omega^j) = \sum_{i=0}^{N-1} a_i \omega^{ij}, \quad j = 0, \dots, N-1.$$

The inverse is interpolation on H , and it has a closed form:

$$\text{NTT}^{-1}(\hat{a})_i = a_i = N^{-1} \sum_{j=0}^{N-1} \hat{a}_j \omega^{-ij}.$$

Remark. The NTT is not a new idea — it is Lagrange interpolation from Section 2, on the special node set H . What the special node set buys is speed: the naive formula costs $O(N^2)$, and the structure of H reduces it to $O(N \log N)$.

6.2 Cooley–Tukey

Split a into even- and odd-indexed coefficients:

$$a(X) = a_e(X^2) + X \cdot a_o(X^2).$$

Since ω^2 has order $N/2$, evaluating a_e and a_o on the subgroup of order $N/2$ is two transforms of half the size. The results are recombined by the **butterfly**

$$\hat{a}_j = a_e(\omega^{2j}) + \omega^j a_o(\omega^{2j}).$$

This gives $T(N) = 2T(N/2) + O(N)$, hence $T(N) = O(N \log N)$.

6.3 A Worked 4-Point Transform

Work in $\mathbb{F}_{17}$ with $N = 4$ and $\omega = 13$, so $(\omega^0, \omega^1, \omega^2, \omega^3) = (1, 13, 16, 4)$. Take

$$a(X) = 1 + 2X + 3X^2 + 4X^3.$$

Directly. Using $13^2 \equiv 16$, $13^3 \equiv 4$, $16 \equiv -1$, $4^2 \equiv 16$, $4^3 \equiv 13$:

$$\begin{aligned} a(1) &= 1 + 2 + 3 + 4 = 10, \\ a(13) &= 1 + 26 + 48 + 16 = 91 \equiv 6, \\ a(16) &= 1 - 2 + 3 - 4 = -2 \equiv 15, \\ a(4) &= 1 + 8 + 48 + 52 = 109 \equiv 7, \end{aligned} \quad \implies \quad \text{NTT}(a) = (10, 6, 15, 7).$$

By Cooley–Tukey. Here $a_e(Y) = 1 + 3Y$ and $a_o(Y) = 2 + 4Y$, evaluated on $\{\omega^0, \omega^2\} = \{1, 16\}$:

$$a_e(1) = 4, \quad a_e(16) = 49 \equiv 15, \quad a_o(1) = 6, \quad a_o(16) = 66 \equiv 15.$$

The butterflies then give

$$\begin{aligned}\hat{a}_0 &= a_e(1) + \omega^0 a_o(1) = 4 + 1 \cdot 6 = 10, & \hat{a}_1 &= a_e(16) + \omega^1 a_o(16) = 15 + 13 \cdot 15 \equiv 6, \\ \hat{a}_2 &= a_e(1) + \omega^2 a_o(1) = 4 + 16 \cdot 6 \equiv 15, & \hat{a}_3 &= a_e(16) + \omega^3 a_o(16) = 15 + 4 \cdot 15 \equiv 7,\end{aligned}$$

which agrees. Four multiplications instead of twelve — a small saving at $N = 4$, a decisive one at $N = 256$ or $N = 2^{16}$.

Inverting. We need $N^{-1} = 4^{-1} \equiv 13$ and $\omega^{-1} = 13^{-1} \equiv 4$ in $\mathbb{F}_{17}$. For instance

$$a_0 = 13 \cdot (10 + 6 + 15 + 7) = 13 \cdot 38 \equiv 13 \cdot 4 = 52 \equiv 1. \checkmark$$

6.4 Why We Care: Fast Multiplication

Multiplying two polynomials directly costs $O(N^2)$ coefficient multiplications. In the evaluation representation it costs N , because evaluation is a ring homomorphism:

$$(a \cdot b)(\omega^j) = a(\omega^j) \cdot b(\omega^j).$$

So the algorithm is: transform, multiply pointwise, transform back.

$$a \cdot b = \text{NTT}^{-1}(\text{NTT}(a) \odot \text{NTT}(b)), \quad O(N \log N) \text{ instead of } O(N^2).$$

Forward reference. Two adjustments are needed before this is usable in Lecture 3. First, the product of two degree- $< N$ polynomials has degree up to $2N - 2$, so a transform of length N computes the product modulo $X^N - 1$ — a *cyclic* convolution. Second, what R_q needs is multiplication modulo $X^n + 1$, the *negacyclic* convolution. The fix is a twist by a primitive $2n$ -th root ψ with $\psi^2 = \omega$: scale $a_i \mapsto \psi^i a_i$ before the transform and undo it afterwards. This is why the standards need $2n \mid q - 1$ rather than merely $n \mid q - 1$, and it is why the FIPS 203 specification defines ML-KEM’s arithmetic directly in the NTT domain.

Summary

1. $\mathbb{F}_p[X]$ has a division algorithm, hence gcds, Bézout, and inverses — Lecture 1 transplanted from $\mathbb{Z}$ to polynomials.
2. A nonzero polynomial of degree d over a field has at most d roots. Nearly every argument in this lecture is a corollary.
3. Coefficients and evaluations at k distinct points are two representations of the same degree- $< k$ polynomial; Lagrange moves between them.
4. Hiding data in the coefficients gives secret sharing; hiding it in the roots gives set intersection.
5. Schwartz–Zippel makes a random evaluation a sound test of polynomial identity, with error $d/|\mathbb{F}|$.
6. On a multiplicative subgroup H of order N , the vanishing polynomial is just $X^N - 1$, and “holds everywhere on H ” becomes “divisible by $X^N - 1$ ”.
7. $\mathbb{F}_p$ has an N -th root of unity iff $N \mid p - 1$. The two-adicity of p decides how far a power-of-two NTT can go.
8. The NTT is interpolation on the roots of unity, computed in $O(N \log N)$; it turns polynomial multiplication into pointwise multiplication.

Exercises

1. Show that $X^2 - 1$ has four roots in $\mathbb{Z}_8$, and identify precisely which step of the proof of Corollary 2 fails there.
2. In $\mathbb{F}_{17}$, use the extended Euclidean algorithm in $\mathbb{F}_{17}[X]$ to find the inverse of $X + 2$ modulo $X^2 + 1$. Verify your answer by multiplying out.
3. Recover the Shamir secret of Section 2.1 using shares $\{2, 4, 5\}$ instead of $\{1, 2, 3\}$, and confirm you obtain 13.
4. Alice's PSI polynomial for the set $\{3, 4, 5\}$ is $60 - 47X + 12X^2 - X^3$. Verify that it vanishes at 3, 4 and 5, and evaluate it at 6. What does Bob learn when his element is 6?
5. A prover claims two degree- 2^{20} polynomials over $\mathbb{F}_p$ are equal, where $p \approx 2^{128}$. Bound the probability that a single random evaluation fails to catch a false claim. How many independent random evaluations would you need over $\mathbb{F}_{17}$ to reach comparable confidence?
6. List the elements of the subgroup of order 8 in $\mathbb{F}_{17}^\times$, and write down its vanishing polynomial. Confirm your answer by checking the sum and product of the elements against the coefficients.
7. Determine the two-adicity of $p = 12289$ (used by Falcon). What is the largest power-of-two NTT this field supports?
8. Compute $\text{NTT}(b)$ for $b(X) = 4 + 3X + 2X^2 + X^3$ over $\mathbb{F}_{17}$ with $\omega = 13$. Then compute the pointwise product with the $\text{NTT}(a)$ of Section 6.3 and apply NTT^{-1} . Which polynomial have you obtained, and why is it *not* $a \cdot b$?
9. Prove that if P vanishes on a subgroup H of order N then $\deg P \geq N$ unless $P = 0$. Where is this used in Section 4.1?